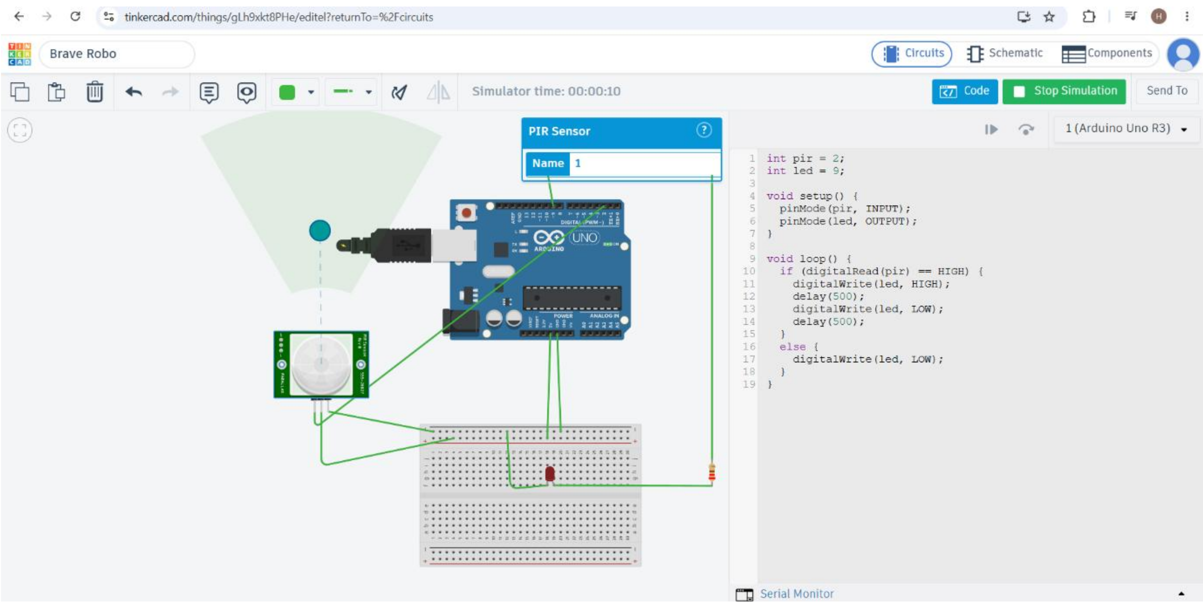
TASK-2

SENSOR BASED SIMULATION



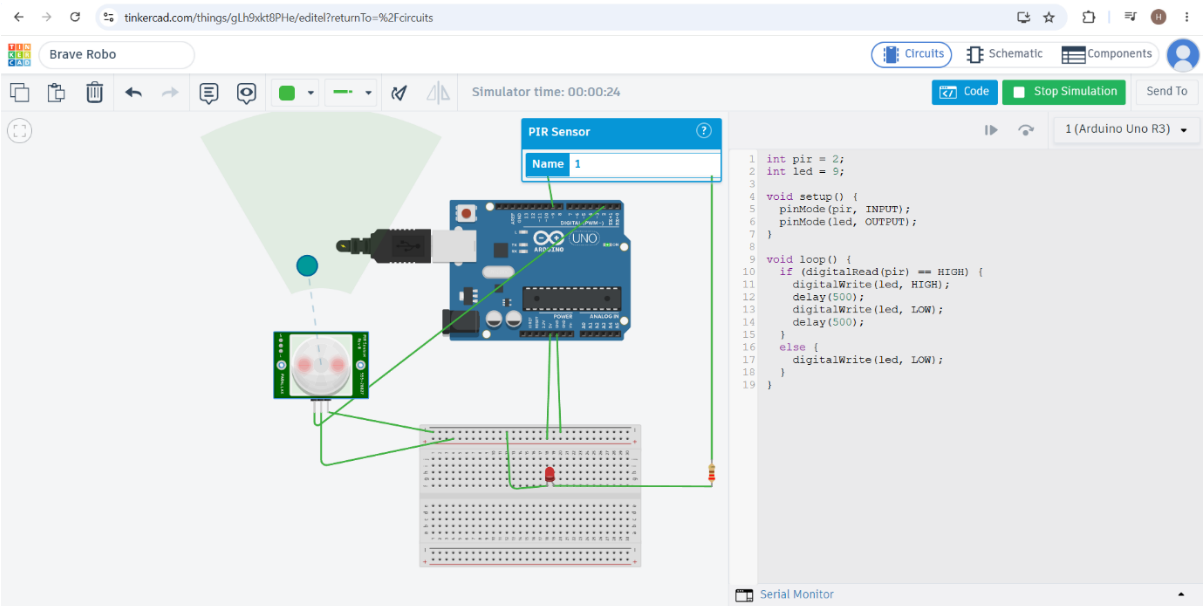


Fig: LED Blinking

1.CODE

```
int pir = 2;
int led = 9;

void setup() {
  pinMode(pir, INPUT);
  pinMode(led, OUTPUT);
}

void loop() {
  if (digitalRead(pir) == HIGH) {
    digitalWrite(led, HIGH);
    delay(500);
    digitalWrite(led, LOW);
    delay(500);
  }
  else {
    digitalWrite(led, LOW);
  }
}
```

2.CODE EXPLANATION – PIR SENSOR AND BLINKING LED

This program is used to detect motion using a PIR sensor and blink an LED.

1. int pir = 2;
   The PIR sensor is connected to digital pin 2 of the Arduino.

2. int led = 9;
   The LED is connected to digital pin 9 of the Arduino.
3. setup() function:
   - pinMode(pir, INPUT);
     Sets the PIR sensor as an input because it sends a signal to the Arduino.
   - pinMode(led, OUTPUT);
     Sets the LED as an output because the Arduino controls the LED.
4. loop() function:
   The Arduino continuously checks the PIR sensor for motion.
5. if (digitalRead(pir) == HIGH):
   If the PIR sensor detects motion, it gives a HIGH signal.
6. digitalWrite(led, HIGH);
   The LED turns ON.
7. delay(500);
   The program waits for 500 milliseconds (0.5 seconds).
8. digitalWrite(led, LOW);
   The LED turns OFF.
9. delay(500);
   The program waits for another 500 milliseconds.
10. else:
    If no motion is detected, the LED remains OFF.

3.WORKING PRINCIPLE

1. The PIR sensor detects movement in its surroundings.
2. When motion is detected, the PIR sensor sends a HIGH signal to the Arduino.
3. The Arduino receives the signal through digital pin 2.
4. The Arduino turns ON the LED connected to digital pin 9.

5. The LED stays ON for 0.5 seconds and then turns OFF.
6. The LED continues blinking while motion is detected.
7. When no motion is detected, the PIR sensor sends a LOW signal.
8. The Arduino keeps the LED OFF when there is no motion.